The Fractional Fourier Transform on Graphs: Modulation and Convolution

Yu Zhang
School of Mathematics and Statistics
Beijing Institute of Technology
Beijing, China
1510895484@qq.com

*Bing-Zhao Li
School of Mathematics and Statistics
Beijing Institute of Technology
Beijing, China
li_bingzhao@bit.edu.cn

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***Abstract*—The fractional Fourier transform on graph (GFrFT) is an extension of discrete fractional Fourier transform (DFrFT) based on graph signal processing (GSP). In this paper, we discuss the shift, polynomial filters, delta functions, the convolution and modulation of graph signal in fractional domain. First, we review the basic theory of discrete graph signal processing, concepts of graph signals and GFrFT. Then, similar to the definition of the shift of adjacency matrix in vertex domain, the shift matrix of graph spectral domain in the GFrFT is defined. Finally, the new polynomial filters are used to illustrate the convolution and modulation of fractional graph signals in the vertex domain and graph spectral domain, and the relationship between the delta functions and convolution is discussed.**

***Index Terms*—Graph signal processing, modulation, convolution, fractional Fourier transform on graph**

I. Introduction

With the rapid increase of information, more and more data are defined on the irregular graph structure. The graph signal processing (GSP) can analyze and interpret these complex and irregular underlying signals and provide a signal processing framework [1], [2]. However, as the difficulty of signal processing increases, many time-frequency signal processing techniques are introduced into graph signals: the graph wavelet transform [3], the windowed graph Fourier transform [4], the fractional Fourier transform on graphs (GFrFT) [5], the linear canonical transform on graphs [6] and the windowed graph fractional Fourier transform [7], etc. At present, the GSP based on fractional Fourier transform is becoming popular and developing continuously. The main content of this paper is to further study the properties of the GFrFT.

The fractional Fourier transform (FrFT) was proposed in the 1980s and introduced into the field of signal processing in the 1990s [8]. It is a generalization of Fourier transform, which rotates the signal to the middle region between time and frequency. To observe the local information of graph signal and get the conversion process from vertex domain to graph spectral domain, the FrFT is extended to GSP, and the GFrFT [5] is proposed. Similar to the GSP, its adjacency matrix is the shift operator of graph signal [9], [10]. On this basis, we define the graph spectral domain shift $\mathbf{S}_a$ under the GFrFT, which is similar to the shift definition of the vertex domain adjacency matrices $\mathbf{A}$ and $\mathbf{A}_a$ [11]–[13]. Using the polynomial filters $H(\mathbf{A}_a)$ and $H(\mathbf{S}_a)$ different from [14], the convolution and modulation of graph signals representing vertices and spectra in the fractional domain are further discussed.

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In this paper, the basic concepts of graph signals and the GFrFT are reviewed in Section II. Then, in Section III, we define shift $\mathbf{S}_a$ in the spectral domain under the GFrFT. Modulation and convolution in GFrFT in Section IV are defined by using the vertex domain shift $\mathbf{A}_a$ and the graph spectral domain shift $\mathbf{S}_a$, and discuss the relationship between the delta functions and convolution. Finally, in Section V, the paper is concluded.

II. The Fractional Fourier Transform on Graphs

In this section, we briefly introduce the basic concepts of graph signals and the definition and notion of the GFrFT.

A. Graph Signals

In discrete signal processing on graphs, the high-dimensional structure of signals are represented by a graph $\mathcal{G} = (\mathcal{V}, \mathbf{A})$, where $\mathcal{V} = v_0, \ldots, v_{N-1}$ denotes the set of vertices and $\mathbf{A} \in \mathbb{C}^{N \times N}$ denotes the weighted adjacency matrix. The graph signals are defined as a mapping that maps the vertex v_n to a signal coefficient $x_n \in \mathbb{C}$, which can also be written as a complex vector

$$x = \begin{bmatrix} x_0 & x_1 & \ldots & x_{N-1} \end{bmatrix}^\top \in \mathbb{C}^N.$$

B. Graph Fractional Fourier Transform

Graph Fourier transform (GFT) [1], [2] is defined by eigen-decomposition of adjacency matrix $\mathbf{A}$ or the Laplace matrix $\mathcal{L}$, here we use the adjacency matrix.

$$\mathbf{A} = \mathbf{V}\mathbf{\Lambda}_{\mathbf{A}}\mathbf{V}^{-1}, \tag{1}$$

where the columns of $\mathbf{V}$ are the eigenvectors of $\mathbf{A}$, corresponding to eigenvalues $\lambda_0, \ldots, \lambda_{N-1}$ in the diagonal matrix $\mathbf{\Lambda}_{\mathbf{A}} \in \mathbb{C}^{N \times N}$.

The convolution in the vertex domain is defined as matrix vector multiplication [10]

$$H(\mathbf{A})v, \tag{2}$$

where $H(\mathbf{A})$ is a shifted polynomial and $\mathbf{A}$ is an adjacency matrix.

The ath order fractional graph Fourier transform of the graph signals [5] $\mathbf{s} \in \mathbb{C}^{N\times 1}$ is defined as

$$\hat{\mathbf{s}} = \mathbf{F}^a \mathbf{s} = \mathbf{Q}\mathbf{\Lambda}^a \mathbf{Q}^{-1}\mathbf{s}, \quad a \in [0,1], \tag{3}$$

and its inverse transform is

$$\mathbf{s} = \mathbf{F}^{-a}\hat{\mathbf{s}} = \mathbf{Q}\mathbf{\Lambda}^{-a}\mathbf{Q}^{-1}\hat{\mathbf{s}}, \tag{4}$$

where the matrices $\mathbf{Q}$ and $\mathbf{\Lambda}$ are given by the spectral decomposition of the GFT matrix $\mathbf{V}^{-1}$. The signal $\mathbf{s}$ on the graph $\mathcal{G}$ has a diagonalizable adjacency matrix $\mathbf{A}$. And the GFT matrix $\mathbf{V}^{-1}$ is orthogonal and diagonalized, then

$$\mathbf{F} = \mathbf{V}^{-1} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1}, \tag{5}$$

It is easy to verify that when $a = 0$, the GFrFT of the graph signal is the signal itself. When $a = 1$, the GFrFT is reduced to the GFT.

III. Spectral Shift in Graph Fractional Domain

Similar to the approach of determining the graph shift in the spectral domain in discrete signal processing (DSP), the graph shift $\mathbf{S}_a$ in the spectral domain under DFrFT is defined in the ring graph.

A. Graph Spectral Domain

From DSP [11], the shift in the frequency domain is defined, which is applied to the graph spectral in GSP, and its matrix

$$\mathbf{S} = \mathbf{V}^{-1}\mathbf{\Lambda}^*_{\mathbf{A}}\mathbf{V}, \tag{6}$$

where $\mathbf{\Lambda}_{\mathbf{A}}$ is the diagonal matrix of eigenvalues of the adjacency matrix $\mathbf{A}$ of an arbitrary graph, and $\mathbf{\Lambda}^*_{\mathbf{A}}$ is the conjugate of $\mathbf{\Lambda}_{\mathbf{A}}$. In general, $\mathbf{S}$ is not equal to $\mathbf{A}$ in the GSP.

The matrix $\mathbf{S}$ is the same as $\mathbf{A}$, and they play the roles of adjacency matrix and shift matrix in the graph spectral domain and vertex domain, respectively. $\mathbf{A}$ can have the same structure as $\mathbf{A}$ but with different weights, further, $\mathbf{A}$ can have a different structure than $\mathbf{S}$.

B. Shift Matrix of Vertex Domain in GFrFT

For the adjacency matrix we have $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}_{\mathbf{A}}\mathbf{V}^{-1}$, and for the fractional Fourier transform matrix having $\mathbf{F}^a = (\mathbf{V}^{-1})^a = \mathbf{Q}\mathbf{\Lambda}^a\mathbf{Q}^{-1}$, therefore, in the vertex domain we construct new shift matrix

$$\mathbf{A}_a = \mathbf{F}^{-a}\mathbf{\Lambda}_{\mathbf{A}}\mathbf{F}^a = \mathbf{Q}\mathbf{\Lambda}^{-a}\mathbf{Q}^{-1}\mathbf{\Lambda}_{\mathbf{A}}\mathbf{Q}\mathbf{\Lambda}^a\mathbf{Q}^{-1}. \tag{7}$$

Reconstructing the shift matrix in the vertex domain, we then reconstruct the shift matrix in the graph spectral domain and polynomial filter.

C. Shift Matrix of Spectral Domain in GFrFT

In any undirected graph, the frequency domain shift has the following property

$$e^{j\frac{2\pi}{N}u_0 k a} x_k \xrightarrow{\mathbf{F}^a} \hat{x}_{u-u_0 a}. \tag{8}$$

The shift $\hat{x}_u$ in frequency is equivalent to the multiplication of the ath order complex exponent in time. And the fractional Fourier component of the shifted graph in a vector

$$(\mathbf{\Lambda}^*_{\mathbf{A}})^{u_0 a}\, x \xrightarrow{\mathbf{F}^a} [\hat{x}_{-u_0 a}, \ldots, \hat{x}_{N-1-u_0 a}]^T,$$

where $\mathbf{\Lambda}^*_{\mathbf{A}} = \mathrm{diag}(e^{j\frac{2\pi}{N}k})$ and similar to the eigendecomposition of the shift matrix $\mathbf{A}$ in the vertex domain of the GSP, we define the shift in the frequency domain of arbitrary graphs of fractional domain

$$\mathbf{S}_a = \mathbf{F}^a\mathbf{\Lambda}^*_{\mathbf{A}}\mathbf{F}^{-a} = \mathbf{Q}\mathbf{\Lambda}^a\mathbf{Q}^{-1}\mathbf{\Lambda}^*_{\mathbf{A}}\mathbf{Q}\mathbf{\Lambda}^{-a}\mathbf{Q}^{-1}. \tag{9}$$

Given a signal $\mathbf{x}$, let $\hat{\mathbf{x}} = \mathbf{Q}\mathbf{\Lambda}^a\mathbf{Q}^{-1}\mathbf{x}$, we shift $\hat{\mathbf{x}}$ through matrix $\mathbf{S}_a$

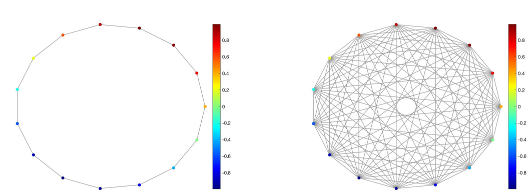
$$\begin{aligned}\mathbf{S}_a\hat{\mathbf{x}} =& \mathbf{Q}\mathbf{\Lambda}^a\mathbf{Q}^{-1}\mathbf{\Lambda}^*_{\mathbf{A}}\mathbf{Q}\mathbf{\Lambda}^{-a}\mathbf{Q}^{-1}\ \hat{\mathbf{x}} \\ =& \mathbf{Q}\mathbf{\Lambda}^a\mathbf{Q}^{-1}\mathbf{\Lambda}^*_{\mathbf{A}}\mathbf{Q}\mathbf{\Lambda}^{-a}\mathbf{Q}^{-1}\ \mathbf{Q}\mathbf{\Lambda}^a\mathbf{Q}^{-1}\mathbf{x} \\ =& \mathbf{Q}\mathbf{\Lambda}^a\mathbf{Q}^{-1}\mathbf{\Lambda}^*_{\mathbf{A}}\mathbf{x},\end{aligned} \tag{10}$$

where $\mathbf{\Lambda}^*_{\mathbf{A}} = \mathrm{diag}(e^{j\frac{2\pi}{N}k})$ is the conjugate matrix of the diagonal matrix corresponding to the eigenvalue of $\mathbf{A}$, and the matrices $\mathbf{Q}$ and $\mathbf{\Lambda}^a$ are given by the spectral decomposition of GFT matrix $\mathbf{V}^{-1}$.

This produces a FrFT pair similar to (7)

$$\mathbf{\Lambda}^*_{\mathbf{A}}\mathbf{x} \xrightarrow{\mathbf{F}^a} \mathbf{S}_a\hat{\mathbf{x}}. \tag{11}$$

In particular, $\mathbf{A}$ is a real valued adjacency matrix and other properties are consistent with those in the ring graph, hence, for ring graph, $\mathbf{A}_a = \mathbf{A}^*_a = \mathbf{S}_a$. Fig. 1. (a) and (b) are ring graphs constructed by matrices in the vertex domain and graph spectral domain, respectively, where $N = 15, \mathbf{s} = \sin(2k\pi/N),\ k = 0, \ldots N-1$.



(a) Shift $\mathbf{A}_a$ (b) Shift $\mathbf{S}_a$

Fig. 1. Ring graph.

Here we take the inverse GFrFT of $\mathbf{S}\hat{\mathbf{x}}$

$$\mathbf{Q}\mathbf{\Lambda}^{-a}\mathbf{Q}^{-1}\mathbf{S}_a\hat{\mathbf{x}} = \mathbf{\Lambda}^*_{\mathbf{A}}\mathbf{x} = \mathrm{diag}(e^{j\frac{2\pi}{N}k})\mathbf{x}. \tag{12}$$

IV. Convolution and Modulation in Graph Fractional Domain

Since convolution in one domain is modulation (multiplication) in another domain, convolution only needs to be considered in vertex domain and graph spectral domain. Before defining convolution and modulation in GSP of fractional domain, we need to redefine the polynomial filter $H(\cdot)$ in the GSP of fractional domain.

A. The Polynomial Filter in Graph Fractional Domain

Since the vertex domain and the graph spectral domain are dual, we first consider the polynomial filter of shift $\mathbf{A}$ in the vertex domain

$$\begin{aligned} H\left(\mathbf{A}\right) =& h_{N-1}\mathbf{A}^{N-1} + h_{N-2}\mathbf{A}^{N-2} + ... + h_1\mathbf{A} + h_0\mathbf{I} \\ =& \mathbf{V}H\left(\mathbf{\Lambda_A}\right)\mathbf{V}^{-1}. \end{aligned} \tag{13}$$

The graph signal $\mathbf{x}$ is input into $H(\mathbf{A})$, and in the graph spectral domain, the graph frequency response modulation (the Hadamard product) $\hat{\mathbf{x}}$ of $\hat{\mathbf{y}}$ is as follows

$$H\left(\mathbf{A}\right)\mathbf{x} \xrightarrow{\mathbf{F}} \hat{\mathbf{y}} \circ \hat{\mathbf{x}}, \tag{14}$$

where $H\left(\mathbf{A}\right)\mathbf{x}$ denotes vertex filtering, $\hat{\mathbf{y}} \circ \hat{\mathbf{x}}$ is spectral modulation and the spectral response $\hat{\mathbf{y}}$ was defined as $H\left(\mathbf{\Lambda_A}\right)$.

However, in the GFrFT, the fractional parameter needs to be introduced, hence the polynomial filter $H\left(\mathbf{A}\right)$ should be reconstructed. First, assume a fractional Fourier transform pair

$$H_a\left(\mathbf{A}\right)\mathbf{x} \xrightarrow{\mathbf{F}^a} \hat{\mathbf{y}} \circ \hat{\mathbf{x}}. \tag{15}$$

Obviously the construction of $H_a\left(\mathbf{A}\right)$ is the construction of $H\left(\mathbf{A}_a\right)$, that is, the reconstruction of the shift matrix $\mathbf{A}$. Then the new polynomial filter is

$$\begin{aligned} H\left(\mathbf{A}_a\right) =& h_{N-1}\mathbf{A}_a^{N-1} + h_{N-2}\mathbf{A}_a^{N-2} + ... + h_1\mathbf{A}_a + h_0\mathbf{I} \\ =& \mathbf{F}^{-a}H\left(\mathbf{\Lambda_A}\right)\mathbf{F}^{a}. \end{aligned} \tag{16}$$

Similarly, combined with III-C, a new polynomial filter in the spectral domain can be constructed

$$\begin{aligned} H\left(\mathbf{S}_a\right) =& h_{N-1}\mathbf{S}_a^{N-1} + h_{N-2}\mathbf{S}_a^{N-2} + ... + h_1\mathbf{S}_a + h_0\mathbf{I} \\ =& \mathbf{F}^{a}H\left(\mathbf{\Lambda}^*_\mathbf{A}\right)\mathbf{F}^{-a}. \end{aligned} \tag{17}$$

B. Convolution of Vertex Domain in GFrFT

The convolution of the vertex domain is filtered by the graph polynomial filter [10], [11], assuming that the convolution can be expressed as follows

$$\mathbf{y} * \mathbf{x} = H\left(\mathbf{A}_a\right)\mathbf{x}. \tag{18}$$

We consider the relationship between vertex filtering and spectral modulation

$$H\left(\mathbf{A}_a\right)\mathbf{x} \xrightarrow{\mathbf{F}^a} \hat{\mathbf{y}} \circ \hat{\mathbf{x}}, \tag{19}$$

where, $\hat{\mathbf{y}}$ and $\hat{\mathbf{x}}$ are graph signals and the convolution (18) can be proved when $\mathbf{F}^a\left(H\left(\mathbf{A}_a\right)\mathbf{x}\right) = \hat{\mathbf{y}} \circ \hat{\mathbf{x}}$.

In the GFrFT, the matrix vector product between matrix $H\left(\mathbf{A}_a\right)$ and vector $\mathbf{x}$ is the Hadamard product of signal $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$. Associate the matrix $H\left(\mathbf{A}_a\right)$ with the signal $\hat{\mathbf{y}}$.

$$\begin{aligned} H\left(\mathbf{A}_a\right)\mathbf{x} =& \mathbf{Q\Lambda}^{-a}\mathbf{Q}^{-1}(h_{N-1}\mathbf{\Lambda_A}^{N-1} + h_{N-2}\mathbf{\Lambda_A}^{N-2} \\ & + ... + h_1\mathbf{\Lambda_A} + h_0\mathbf{I})\mathbf{Q\Lambda}^{a}\mathbf{Q}^{-1}\mathbf{x} \\ =& \mathbf{Q\Lambda}^{-a}\mathbf{Q}^{-1}(h_{N-1}\mathbf{\Lambda_A}^{N-1} + h_{N-2}\mathbf{\Lambda_A}^{N-2} \\ & + ... + h_1\mathbf{\Lambda_A} + h_0\mathbf{I})\hat{\mathbf{x}} \\ =& \mathbf{Q\Lambda}^{-a}\mathbf{Q}^{-1}H\left(\mathbf{\Lambda_A}\right)\hat{\mathbf{x}} \\ =& \mathbf{F}^{-a}H\left(\mathbf{\Lambda_A}\right)\hat{\mathbf{x}}. \end{aligned}$$

Thus, we take the GFrFT of $H\left(\mathbf{A}_a\right)\mathbf{x}$ and get

$$\mathbf{F}^a\left(H\left(\mathbf{A}_a\right)\mathbf{x}\right) = H\left(\mathbf{\Lambda_A}\right)\hat{\mathbf{x}} = \mathbf{F}^a\mathbf{y} \circ \mathbf{F}^a\mathbf{x} = \hat{\mathbf{y}} \circ \hat{\mathbf{x}}, \tag{20}$$

where $\mathbf{\Lambda_A}$ is a diagonal matrix, thus $H\left(\mathbf{\Lambda_A}\right)$ is also diagonal. Thus, $H\left(\mathbf{\Lambda_A}\right) = \mathrm{diag}\left(\hat{\mathbf{y}}\right)$, and the matrix $\mathrm{diag}\left(\hat{\mathbf{y}}\right)$ is the graph frequency response of polynomial filter $H\left(\mathbf{A}_a\right)$.

Therefore, we have

$$H\left(\mathbf{A}_a\right) = \mathbf{F}^{-a}\mathrm{diag}\left(\hat{\mathbf{y}}\right)\mathbf{F}^{a}, \tag{21}$$

and the Hadamard product of signal $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ in the spectral domain is the matrix vector product of matrix $H\left(\mathbf{A}_a\right)$ and signal $\hat{\mathbf{x}}$.

C. Convolution of Spectral Domain in GFrFT

The convolution of two graph signals $\hat{\mathbf{y}}$ and $\hat{\mathbf{x}}$ is filtered in the spectral domain by the spectrogram polynomial filter $H\left(\mathbf{S}_a\right)$ given by (17), where $\mathbf{S}_a$ is the new spectral shift, and the convolution can also be assumed as follows

$$\hat{\mathbf{y}} * \hat{\mathbf{x}} = H\left(\mathbf{S}_a\right)\hat{\mathbf{x}}. \tag{22}$$

Since the two domains are dual, we consider the relationship between spectral filtering and vertex modulation in graph spectral domain similar to convolution in vertex domain

$$\mathbf{y} \circ \mathbf{x} \xrightarrow{\mathbf{F}^a} H\left(\mathbf{S}_a\right)\hat{\mathbf{x}}, \tag{23}$$

where, $\mathbf{y}$ and $\mathbf{x}$ are graph signals and the convolution (22) can be proved when $\mathbf{F}^{-a}\left(H\left(\mathbf{S}_a\right)\hat{\mathbf{x}}\right) = \mathbf{y} \circ \mathbf{x}$.

In the GFrFT, the Hadamard product of signal $\mathbf{x}$ and $\mathbf{y}$ is the matrix vector product between matrix $H\left(\mathbf{S}_a\right)$ and vector $\hat{\mathbf{x}}$. Associate the matrix $H\left(\mathbf{S}_a\right)$ with the signal $\hat{\mathbf{y}}$. Similarly relate the matrix $H\left(\mathbf{S}_a\right)$ to the signal $\mathbf{y}$.

$$\begin{aligned} H\left(\mathbf{S}_a\right)\hat{\mathbf{x}} =& \mathbf{Q\Lambda}^{a}\mathbf{Q}^{-1}(h_{N-1}\mathbf{\Lambda}^*_\mathbf{A}{}^{N-1} + h_{N-2}\mathbf{\Lambda}^*_\mathbf{A}{}^{N-2} \\ & + ... + h_1\mathbf{\Lambda}^*_\mathbf{A} + h_0\mathbf{I})\mathbf{Q\Lambda}^{-a}\mathbf{Q}^{-1}\hat{\mathbf{x}} \\ =& \mathbf{Q\Lambda}^{a}\mathbf{Q}^{-1}(h_{N-1}\mathbf{\Lambda}^*_\mathbf{A}{}^{N-1} + h_{N-2}\mathbf{\Lambda}^*_\mathbf{A}{}^{N-2} \\ & + ... + h_1\mathbf{\Lambda}^*_\mathbf{A} + h_0\mathbf{I})\mathbf{x} \\ =& \mathbf{Q\Lambda}^{a}\mathbf{Q}^{-1}H\left(\mathbf{\Lambda}^*_\mathbf{A}\right)\mathbf{x} \\ =& \mathbf{F}^{a}H\left(\mathbf{\Lambda}^*_\mathbf{A}\right)\mathbf{x}. \end{aligned}$$

Thus, we get

$$\mathbf{F}^{-a}\left(H\left(\mathbf{S}_a\right)\hat{\mathbf{x}}\right) = H\left(\mathbf{\Lambda}^*_\mathbf{A}\right)\mathbf{x} = \mathbf{F}^{-a}\hat{\mathbf{y}} \circ \mathbf{F}^{-a}\hat{\mathbf{x}} = \mathbf{y} \circ \mathbf{x}. \tag{24}$$

where $\mathbf{\Lambda}^*_\mathbf{A}$ is a diagonal matrix, so $H\left(\mathbf{\Lambda}^*_\mathbf{A}\right)$ is also diagonal. Thus, $H\left(\mathbf{\Lambda}^*_\mathbf{A}\right) = \mathrm{diag}\left(\mathbf{y}\right)$.

Therefore, we have

$$H\left(\mathbf{S}_a\right)=\mathbf{F}^a \operatorname{diag}(\mathbf{y}) \mathbf{F}^{-a}, \tag{25}$$

and the matrix vector product of matrix $H\left(\mathbf{S}_a\right)$ and signal $\mathbf{x}$ is the Hadamard product of signal $\mathbf{x}$ and $\mathbf{y}$ in the vertex domain.

D. Delta Functions in Graph Fractional Domain

We consider the δ function and its shifts in GSP of fractional domain.

$$\delta_i[n]=\begin{cases}1, & n=i, \\ 0, & \text{otherwise.}\end{cases} \tag{26}$$

In DSP of the fractional domain, the δ function is impulsive in the time domain and flat in the frequency domain, respectively $\delta_0=[1\ 0 \ldots 0]^{\top}$ and $\hat{\delta}_0=(1/\sqrt{N})[1\ 1 \ldots 1]^{\top}$. The DSP nth time shifted δ is

$$\delta_n=\mathbf{A}_a^n \cdot \delta_0=[0\ 0 \cdots 1]^{\top}. \tag{27}$$

Thus

$$\hat{\delta}_n=\frac{1}{\sqrt{N}} \lambda^n=\frac{1}{\sqrt{N}}\begin{bmatrix}\lambda_0^n \\ \lambda_1^n \\ \vdots \\ \lambda_{N-1}^n\end{bmatrix}=\frac{1}{\sqrt{N}}\begin{bmatrix}1 \\ e^{-j \frac{2 \pi}{N} n} \\ \vdots \\ e^{-j \frac{2 \pi}{N}(N-1) n}\end{bmatrix}, \tag{28}$$

where the entries of the vector λ^n are the n powers of the graph frequencies. The DSP nth spectral shifted δ is

$$\hat{\delta}_n=\mathbf{S}_a^n \cdot \hat{\delta}_0=\mathbf{F}^a \mathbf{\Lambda}_{\mathbf{A}}^{*^n} \mathbf{F}^{-a} \hat{\delta}_0=\mathbf{F}^a \mathbf{\Lambda}_{\mathbf{A}}^{*^n} \delta_0 \stackrel{(i)}{=} \mathbf{F}^a \delta_0=\hat{\delta}_0, \tag{29}$$

where, $\mathbf{F}^a=\mathbf{Q \Lambda Q^{-1}}$ and equation (i) since the first diagonal term of $\mathbf{\Lambda}_{\mathbf{A}}^{*^n}$ is 1, multiplication by the delta function is still a delta function. It can be seen that $\hat{\delta}_0=\hat{\delta}_n$ and cyclic shifts of the flat $\hat{\delta}_0$ obtains the same function.

In GSP of the fractional domain, similar to DSP in the fractional domain, the delta function is impulsive in the vertex domain and flat in the graph spectral domain. We first consider the delta function in the vertex domain.

1) Delta Functions in Vertex Domain:

$$\begin{aligned} \hat{\delta}_0 & =\mathbf{F}^a \cdot[1\ 0\ \cdots\ 0]^{\top}=\mathbf{y}_0, \\ \delta_n & =\mathbf{A}_a^n \cdot \delta_0=d_0, \end{aligned} \tag{30}$$

where $\mathbf{y}_0$ is the first column of $\mathbf{F}^a$ and d_0 is the 0th column of $\mathbf{A}_a^n$.

Hence

$$\begin{aligned} \hat{\delta}_n=\widehat{\mathbf{A}_a^n \cdot \delta_0} & =\mathbf{F}^a \mathbf{F}^{-a} \mathbf{\Lambda}_{\mathbf{A}}^n \mathbf{F}^a \cdot \delta_0 \\ & =\mathbf{\Lambda}_{\mathbf{A}}^n \cdot \mathbf{y}_0=\operatorname{diag}\left[\mathbf{y}_0\right] \cdot \lambda^n, \end{aligned} \tag{31}$$

where $\lambda^n=\left[\lambda_0^n\ \cdots\ \lambda_{N-1}^n\right]^{\top}$ and then,

$$\delta_n=\mathbf{F}^{-a}\left(\mathbf{\Lambda}_{\mathbf{A}}^n \cdot \mathbf{y}_0\right)=\mathbf{F}^{-a}\left(\operatorname{diag}\left[\mathbf{y}_0\right] \lambda^n\right), \tag{32}$$

$$\mathbf{F}^{-a}\left(\mathbf{S}_a^n \cdot \hat{\delta}_0\right)=\mathbf{\Lambda}_{\mathbf{A}}^{*^n}[1\ 0\ \cdots 0]^{\top}=\lambda_0^{*^n}[1\ 0\ \cdots\ 0]^{\top}, \tag{33}$$

Since both $\mathbf{\Lambda}_{\mathbf{A}}^n$ and $\operatorname{diag}\left[\mathbf{y}_0\right]$ are diagonal, $\mathbf{\Lambda}_{\mathbf{A}}^n \cdot \mathbf{y}_0=\operatorname{diag}\left[\mathbf{y}_0\right] \cdot \lambda^n=\mathbf{\Lambda}_{\mathbf{A}}^n \operatorname{diag}\left[\mathbf{y}_0\right] \cdot \mathbf{1}$.

Define an impulse matrix $\mathbf{D}$ and its $\widehat{\mathbf{D}}$, using (30) and (31), we get

$$\mathbf{D} \triangleq\left[\delta_0\ \delta_1\ \cdots\ \delta_{N-1}\right]=\left[d_0^0\ d_0^1\ \cdots\ d_0^{N-1}\right], \tag{34}$$

$$\begin{aligned} \widehat{\mathbf{D}} & =\mathbf{F}^a \cdot \mathbf{D} \\ & =\frac{1}{\sqrt{N}}\left[\hat{\delta}_0\ \hat{\delta}_1\ \cdots\ \hat{\delta}_{N-1}\right] \\ & =\frac{1}{\sqrt{N}} \operatorname{diag}\left[\mathbf{y}_0\right]\left[\lambda^0\ \lambda^1\ \cdots\ \lambda^{N-1}\right] \\ & =\operatorname{diag}\left[\mathbf{y}_0\right] \frac{1}{\sqrt{N}}\begin{bmatrix}1 & \lambda_0 & \cdots & \lambda_0^{N-1} \\ \vdots & & & \vdots \\ 1 & \lambda_0^{N-1} & \cdots & \lambda_{N-1}^{N-1}\end{bmatrix}. \end{aligned} \tag{35}$$

We apply the assumption in [11] that all entries of $\mathbf{y}_0$ are nonzero and the shift matrix $\mathbf{A}_a$ has different eigenvalues, then under two assumptions, $\mathbf{D}$ and $\widehat{\mathbf{D}}$ are full rank invertible. For DSP of the fractional domain, $\mathbf{y}_0=\frac{1}{\sqrt{N}} \mathbf{1}, \operatorname{diag}\left[\mathbf{y}_0\right]=\frac{1}{\sqrt{N}} \mathbf{I}$, where $\mathbf{I}$ is the identity matrix, normalize the matrix $\widehat{\mathbf{D}}$, get the matrix $\mathcal{D}$ (Vandermonde matrix).

$$\mathcal{D}=\frac{1}{\sqrt{N}}\begin{bmatrix}1 & e^{-j \frac{2 \pi}{N}} & \cdots & e^{-j \frac{2 \pi}{N}(N-1)} \\ \vdots & & & \vdots \\ 1 & e^{-j \frac{2 \pi}{N}(N-1)} & \cdots & e^{-j \frac{2 \pi}{N}(N-1)(N-1)}\end{bmatrix}. \tag{36}$$

2) Delta Functions in Spectral Domain:

$$\hat{\delta}_0=\frac{1}{\sqrt{N}} \mathbf{1}, \delta_n=\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} u_k, \tag{37}$$

where

$$u_k=\frac{1}{\sqrt{N}}\left[1\ e^{j \frac{2 \pi}{N} k}\ \cdots\ e^{j \frac{2 \pi}{N}(n-1)}\right]^{\top},\ k=0, \ldots, N-1.$$

Hence

$$\begin{aligned} \hat{\delta}_n & =\widehat{\mathbf{A}_a^n \cdot \delta_0}=\mathbf{\Lambda}_{\mathbf{A}}^n \cdot \frac{1}{\sqrt{N}} \mathbf{1}=\frac{1}{\sqrt{N}} \lambda^n \\ \delta_n & =\frac{1}{\sqrt{N}} \mathbf{F}^{-a}\left(\lambda^n\right) \end{aligned}. \tag{38}$$

$$\mathbf{F}^{-a}\left(\mathbf{S}_a^n \cdot \hat{\delta}_0\right)=\mathbf{\Lambda}_{\mathbf{A}}^{*^n} \delta_0=\frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} \lambda^n \circ u_i, \tag{39}$$

Define the impulse matrix $\mathbf{D}$ of δ_0 in the graph spectral domain and shift in the vertex domain: for this definition of the graph impulse, only the shift matrix $\mathbf{A}_a$ has different eigenvalues, $\mathbf{D}$ and $\widehat{\mathbf{D}}$ are invertible. It can be obtained by normalizing it.

The result of the delta functions under the fractional order of the two domains are consistent with the shifting delta function in GSP.

E. Relationship between Convolution and Delta Functions

For convolution (18), consider how $H(\mathbf{A}_a)$ is determined from its impulse response $\mathbf{y}$. $H(\mathbf{A}_a)$ has been given in (16). Graph impulse $\mathbf{y}$ is the response of the filter $H(\mathbf{A}_a)$ to the impulse δ_0. Therefore, the relationship between them is (Graph Impulse Response)

$$\begin{aligned}\mathbf{y} =&H\left(\mathbf{A}_a\right)\cdot\delta_0\\ =&h_{N-1}\mathbf{A}_a^{N-1}\delta_0+h_{N-2}\mathbf{A}_a^{N-2}\delta_0+...+h_1\mathbf{A}_a\delta_0+h_0\mathbf{I}\delta_0\\ =&h_{N-1}\delta_{N-1}+h_{N-2}\delta_{N-2}+...+h_1\delta_1+h_0\delta_0\\ =&\mathbf{D}\cdot h,\end{aligned}\tag{40}$$

where the unknown vector of the polynomial filter coefficient is $h=[h_0\ \cdots\ h_{N-1}]^\top$ and $\mathbf{D}$ is the impulse matrix of collecting impulses and its shift, such as (34).

To determine h, the equation (40) can take GFrFT on both sides

$$\hat{\mathbf{y}}=\mathcal{D}\cdot h,\tag{41}$$

here only need to satisfy the shift matrix $\mathbf{A}_a$ has different eigenvalues, the unique h and $H(\mathbf{A}_a)$ can be determined.

On the other hand, the impulse $\hat{\delta}_0$ is introduced in the spectral domain, so that the $\hat{\delta}_0=[1\ \cdots\ 0]^\top$ is impulsive, and $\delta_0=\frac{1}{\sqrt{N}}\mathbf{1}$ is flat. Convolution in the spectral domain, and solve the vector h of the $H(\mathbf{S}_a)$ coefficient by using $\hat{\mathbf{y}}$ as a impulse response (Spectral Filter Graph Impulse Response)

$$\begin{aligned}\hat{\mathbf{y}} =&H\left(\mathbf{S}_a\right)\cdot\hat{\delta}_0\\ =&h_{N-1}\mathbf{S}_a^{N-1}\hat{\delta}_0+h_{N-2}\mathbf{S}_a^{N-2}\hat{\delta}_0+...+h_1\mathbf{S}_a\hat{\delta}_0+h_0\mathbf{I}\hat{\delta}_0\\ =&h_{N-1}\hat{\delta}_{N-1}+h_{N-2}\hat{\delta}_{N-2}+...+h_1\hat{\delta}_1+h_0\hat{\delta}_0\\ =&\mathbf{E}\cdot h,\end{aligned}\tag{42}$$

where the matrix $\mathbf{E}$ is a combination of impulse δ_0 and its shifts.

F. Example

In a cycle graph, let $N=4,\mathbf{x}=\mathbf{y}=[1,1,1,1]^\top,\hat{\mathbf{y}}=\mathbf{F}^a\mathbf{y}=[1.25+0.43i,0.75-0.43i,0.75-0.43i,0.74-0.43i]^\top$, where $a=0.5$. Its adjacency matrix is

$$\mathbf{A}=\begin{bmatrix}0&1&0&1\\1&0&1&0\\0&1&0&1\\1&0&1&0\end{bmatrix}.$$

The polynomial $H\left(\mathbf{A}_a\right)$ can be calculated from Eq. (21), then

$$\mathbf{x}*\mathbf{y}=H\left(\mathbf{A}_a\right)\mathbf{x}=\begin{bmatrix}0.6875+0.7578i\\1.0625-0.3248i\\1.0625-0.3248i\\1.0625-0.3248i\end{bmatrix}=\mathbf{F}^{-a}\left(\hat{\mathbf{y}}\circ\hat{\mathbf{x}}\right).$$

V. Conclusion

In this paper, the shift, the polynomial filter, delta functions, convolution and modulation of graph signals in fractional domain are discussed. Similar to the shift definition of the adjacency matrix $\mathbf{A}_a$ of the vertex domain, the shift $\mathbf{S}_a$ of the spectral domain in the GFrFT is defined. Then, using $\mathbf{A}_a$ and $\mathbf{S}_a$ and the new polynomial filters $H(\mathbf{A}_a)$ and $H(\mathbf{S}_a)$ explains the convolution and modulation of vertex and spectral in the graph fractional domain. Finally, the relationship between the delta function and convolution is discussed, and an example is shown. We hope to provide some ideas and theoretical basis for some future work, such as sampling and reconstruction in the graph fractional domain.

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